

Course: B. Pharm.

Semester: III

Subject with Subject Code: Pharmaceutical Microbiology (BP303T)

Date: 22/12/2018

Marks: 75

Duration: 3hrs

- Instructions:** i) All questions are compulsory
ii) Figures to the right indicate full marks
iii) Draw the diagrams or flow charts wherever necessary.

Q. 1 Choose the correct alternatives.

20

1. In the Riedel walker test strain used is:
 - A. *E. coli*
 - B. *Salmonella typhi*
 - C. *Clostridium tetani*
 - D. *Streptococcus pyrogenous*
2. The value that shows time to kill 90% of microorganisms present in sample is called
 - A. F-value
 - B. T-value
 - C. Z-value
 - D. D-value
3. Biological indicator for radiation sterilisation is
 - A. *B. subtilis*
 - B. *B. pumilus*
 - C. *S. marcescens*
 - D. *B. stearothermophilus*
4. Lysol is a...
 - A. Disinfectant
 - B. Sterilent
 - C. Antiseptic
 - D. Antifungal
5. Which of the following method is best to sterilize heat labile solutions?
 - A. Hot Air Oven
 - B. Autoclave
 - C. Pasteurization
 - D. Membrane filtration
6. Temperature range for 'pasteurization' is
 - A. 60°-70°C
 - B. 62°-72°C
 - C. 65°-75°C
 - D. 121°-130°C
7. Flagella are made up of which protein:
 - A. Flagettetin
 - B. Flagellin
 - C. Flagellicine
 - D. Flagellinocine
8. Viruses are best grown in
 - A. Blood agar
 - B. Enriched media
 - C. Liquid media with serum
 - D. Media with living things

9. Efficiency of HEPA filter is _____ %
- 90.97
 - 88.87
 - 98.79
 - 99.97
10. Test micro organism used for microbiological assay for vitamin B12
- L. viridescens*
 - L. casei*
 - L. leichmanii*
 - None of the above
11. The order of reagents used in the Gram stain are:
- Alcohol, crystal violet, iodine, saffranin
 - Crystal violet, iodine, saffranin, alcohol
 - Crystal violet, iodine, alcohol, saffranin
 - Iodine, crystal violet, saffranin, alcohol
12. Production of acetoin can be detected by which of the following tests?
- Citrate test
 - Voges-Proskauer test
 - Methyl red test
 - Indole test
13. What type of microscopy is usually necessary to observe viruses?
- Dark field
 - Compound
 - Phase contrast
 - Electron
14. Nearly all plant virus are:
- DNA virus
 - RNA virus
 - Viroids
 - Satellite virus
15. When rod shaped bacteria appears to be in pair arrangement it is termed as:
- Staphylococci
 - Bacillus
 - Diplobacilli
 - Streptobacilli
16. Bacteria are sensitive to antibiotics at which phase of growth curve?
- Lag phase
 - Log phase
 - Stationary phase
 - Declined phase
17. Most important surface active disinfectant are
- Anionic
 - Cationic
 - Non-ionic
 - Amphoteric
18. A pH below _____ inhibits animal cell growth
- 7.2
 - 7.0
 - 6.8
 - 7.4

19. Prions are
- Bacteria
 - Rickettsia
 - Viruses
 - Infectious protein
20. Lipopolysaccharides form a part of cell wall of
- gram positive bacteria
 - gram negative bacteria
 - coccus bacteria
 - all of the above

Q. 2 Answer **any two** of the following questions.

20

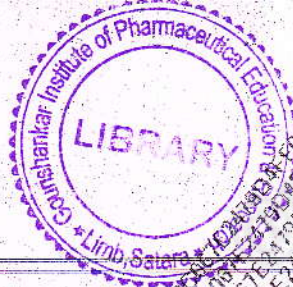
- What are main sources of contamination of an aseptic area? How will you prevent it?
- How will you assess microbial spoilage contamination in pharmaceuticals?
- How will you calculate time require for generation of Bacteria? Explain growth curve of bacteria.

Q. 3 Answer **any seven** of the following questions.

35

- Write a note on cultivation of viruses.
- Write about Bacterial cell.
- Give classification of disinfectants.
- What is Microbiology? Write about different applied branches of Microbiology.
- Write a note on IMViC.
- Write note on chemical indicators.
- Write in detail about the applications of animal cell culture in pharmaceutical industry & research.
- Explain filtration sterilisation.
- Explain any two methods for evaluation of disinfectants.

End



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY

LONERE - RAIGAD - 402 103

Winter Semester Examination - Dec- 2019

Branch: B. Pharm.

Semester: III

Subject: Pharmaceutical Microbiology

Subject Code: BP 303T

Marks: 75

Date: 21/12/2019

Time: 3hrs

- Instructions:**
- i) All questions are compulsory
 - ii) Figures to the right indicate full marks
 - iii) Draw the diagrams or flow charts wherever necessary.

Q. No.1 Multiple Choice Questions (MCQs) = 20 x 1 = 20
(Answer all the questions)

01. Who is Father of Medical Microbiology?

- a) Louis Pasteur b) Alexander Fleming c) Dugger d) None of the above

02. _____ is mainly present in prokaryotic cell

- a) Mitochondria b) Endoplasmic reticulum c) Golgi apparatus d) Mesosomes

03. A binocular microscope has _____

- a) Two objectives b) Two eyepieces c) Two condenser d) Two mirror

04. Phycology is study of _____

- a) Algae b) Fungi c) Molds d) Protozoa

05. In culture collection, NCIB stands for

- a) National Collection of Industrial Bacteria b) National Centre of Industrial Bacteria
c) National Centre of Imported Bacteria d) National Culture of Imported Bacteria

06. Microorganism in natural environment usually occur as _____ culture

- a) Bacterial b) Pure c) Mixed d) Fungal

07. The stain mainly used negative staining

- a) Nigrosin b) Crystal Violet c) Methylene blue d) ZNCF

08. Bacterial cell wall is made up of

- a) chitin b) cellulose c) dextran d) peptidoglycan

09. Pasteurization is carried out at

- a) 100°C b) >100°C c) At 100°C d) None of these

10. Red heat used for sterilization of

- a) Glass slide b) Glass syringe c) Surgical dressing d) Nichrome wire loop

11. Viruses are

- a) Obligate parasites b) free living c) both a and b d) none of these

12. Viruses that attacks bacteria are called

- a) lysophage b) bacteriophage c) virophage d) none of these

13. Most important surface active disinfectant are

- a) Anionic b) Cationic c) Non-ionic d) Amphoteric

14. Efficiency of HEPA filter is _____

- a) 99.97 b) 88.87 c) 90.97 d) 98.79

15. DOP test is used for validation of _____

- a) HEPA filter b) Membrane filter c) Aseptic room d) Autoclave

16. Test microorganism used for microbial assay of streptomycin is

- a) Bacillus pumilus b) *Streptococcus aureus* c) *Bacillus subtilis* d) none of these

17. Dimorphism is characteristic of _____

- a) Bacteria b) Algae c) Viruses d) Fungi

18. Mordant used in gram's staining is

- a) Safranin b) ethanol c) crystal violet d) gram's iodine

19. The organ of locomotion of bacteria is

- a) flagella b) slime c) capsule d) fimbriae

20. Which part of animal virus develops from host cell membrane?

- a) helical b) envelop c) DNA d) RNA

Q. No.2 Long Answers (Answer 2 out of 3) = $2 \times 10 = 20$

- A] Classify sterilization methods and explain in detail moist heat sterilization
- B] Give the classification of disinfectants and write a note on evaluation of disinfectant
- C] Explain in detail structure of bacteria

Q. No.3 Short Answers (Answer 7 out of 9) = $7 \times 5 = 35$

- A] Explain growth curve of Bacteria
- B] Write scope & applications of pharmaceutical Microbiology
- C] Write note on IMViC test
- D] Enlist & explain different type of culture media
- E] What are the sources of contamination in an aseptic area? Give methods of prevention
- F] Explain lytic cycle in detail
- G] Explain the assay of Vitamin B₁₂
- H] Enlist and explain pure culture techniques
- I] Enlist types of spoilage and give the factors affecting microbial spoilage

— END OF PAPER —



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2022

Date: 15.02.2023

Course: B. Pharmacy

Sem: III

Subject Name: Pharmaceutical Microbiology

Subject Code BP303T

Max Marks: 75

Duration:

3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) Differentiate between gram-positive and gram-negative bacteria.
- ii) Write about bacterial growth curve.
- iii) Classify bacterial staining.
- iv) Define i) Sterilization ii) D value
- v) List factors influencing disinfection
- vi) List general properties of viruses
- vii) Differentiate between optical and electron microscope
- viii) Classify clean area as per WHO.
- ix) List ideal characteristics required for preservatives used in pharmaceutical products
- x) Classify methods used for measurement of bacterial growth

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

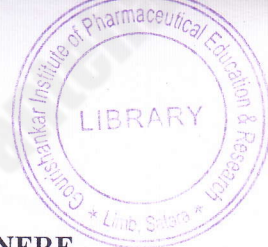
- i) Discuss main sources of contamination in aseptic area? How will you prevent it?
- ii) Classify methods of sterilization and explain in detail moist heat sterilization.
- iii) Define microbiology. Write about various branches of microbiology. Add a note its scope and importance in pharmacy

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) Explain ultra-structure of bacterial cell with suitable diagram.
- ii) Explain IMViC test.
- iii) Give the in-detail classification of disinfectants.
- iv) Explain Phenol coefficient test along with its advantages and disadvantages.
- v) Explain lytic cycle and lysogeny of viruses
- vi) Describe microbial assay of antibiotics.
- vii) Explain tests used for assessment of microbial spoilage
- viii) Write the detail about the applications of animal cell culture in pharmaceutical industry and research.
- ix) Enlist types of spoilage and give the factors affecting microbial spoilage.

-----END OF THE PAPER-----



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2022

Course	: B. Pharmacy	Date: 25/07/2023
Subject Name	: Pharmaceutical Microbiology	Sem: III
Max Marks	: 75	Subject Code : BP303T
		Duration : 3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) Define Microbiology Enlist different branches of microbiology
- ii) Classify Microorganisms
- iii) Classify Bacteria
- iv) Define Industrial Microbiology
- v) Define the term Resolving Power and Numerical Aperture
- vi) Write The Classification of Microscope
- vii) Write A Note On Shape And Arrangement Of Bacteria
- viii) Differentiate Flagella And Pili
- ix) Write The Properties of Agar
- x) Differentiate between total count and viable count

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) Explain in detail applications of Pharmaceutical Microbiology
- ii) Write In Detail About Contribution Of Louis Pasteur In The Field Of Microbiology
- iii) Write In Detail About Discovery Of Antibiotics

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) Differentiate Between Prokaryotic And Eukaryotic Cell
- ii) Write general characteristics of rickettsia.
- iii) Explain the different natural sources of actinomycetes.
- iv) What is dermatophytes.Explain?
- v) Explain in short classification of Fungi.
- vi) How can human amoebiasis be prevented and treated
- vii) Write The Importance Of Algae.
- viii) Write general characteristics of interferon's
- ix) What Are Coliforms And Explain It.

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2023

Date: 06.01.2024

Course: B. Pharmacy
Subject Name: Pharmaceutical Microbiology
Max Marks: 75

Sem: III
Subject Code-BP303T
Duration:

3 Hrs.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) =

- i) Draw a well labeled diagram of Laminar Air Flow equipment.
- ii) Differentiate between prokaryotic & eukaryotic cell.
- iii) List the methods used for evaluation of disinfectants.
- iv) Classify disinfectants.
- v) Define Microbiology. Enlist different branches of Microbiology.
- vi) List various types sterilization monitors.
- vii) Classify the viruses
- viii) Write the morphological classification of bacteria.
- ix) Classify staining techniques.
- x) Differentiate between bacteria and viruses

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) =

- i) Discuss different methods used for measurement of bacterial growth.
- ii) Discuss main sources of contamination in aseptic area and methods of prevention.
- iii) Classify the methods of sterilization and explain heat related methods.

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) =

- i) Define pure culture and explain methods used for isolation of pure culture.
- ii) Explain scope & importance of Microbiology related to Pharmaceutical field.
- iii) Explain bacterial growth curve.
- iv) Discuss methods used for microbiological assay of antibiotics.
- v) Explain designing of aseptic area.
- vi) Explain properties of fungi and add a note on classification of fungi
- vii) Explain in detail IMViC test.
- viii) Explain lytic cycle of bacteriophages in detail.
- ix) Enlist types of spoilage and explain factors affecting microbial spoilage.

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2024



Course: B. Pharmacy

Subject Name: Pharmaceutical Microbiology

Max Marks: 75

Date: 18-06-2024

Semester: III

Subject Code: BP303T

Duration: 3 Hr.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1. Objective Type Questions (Answer All the Questions)

(10X2) =20

- i) Write importance of microbiology
- ii) Differentiate between prokaryotes and eukaryotes
- iii) Define aerobic and anaerobic bacteria
- iv) Define D value and Z value
- v) Classify disinfectant
- vi) Draw a well labeled diagram of laminar air flow
- vii) Write about growth curve phase of bacteria
- viii) Define bacteriostatic and bactericidal
- ix) Write importance of fungi
- x) Write characteristics of viruses

Q.2. Long Answers (Answer 2 out of 3)

(10X2) =20

- i) Explain in details about the structure of bacterial cell
- ii) Define sterilization. Explain different methods of sterilization
- iii) Explain in details about the lytic cycle of bacteriophage

Q.3. Short Answers (Answer 7 out of 9)

(5 X 7) = 35

- i) Discuss about contribution of Louis Pasteur
- ii) Explain simple and gram staining technique
- iii) Explain in details about sterility testing
- iv) Discuss the different sources of contamination in an aseptic area
- v) Explain factors affecting the microbial spoilage of pharmaceutical products
- vi) Discuss about IMViC test
- vii) Write application of animal cell culture
- viii) Write principle and methods of microbiological assay of antibiotics
- ix) Explain in details about preservation of pharmaceutical products

*** END OF THE PAPER ***